

Project 09: Reasoning Model Benchmarking Suite

Empirical Evaluation of LLM Prompt Strategies

1. Executive Overview

The Reasoning Model Benchmarking Suite systematically evaluates model performance across four distinct prompting paradigms: Zero-shot, Few-shot, Chain-of-Thought (CoT), and ReAct. It computes empirical accuracy, average TTFT latency, total token consumption, and cost trade-offs.

2. System Architecture & Workflow Pipeline

- Benchmark Dataset: Math, Logic, Coding & Reasoning Problems
- Evaluator Engine: Executes Zero-shot, Few-shot, CoT, and ReAct paradigms
- Metrics Aggregator: Pass Rate %, TTFT Latency (ms), Token Usage
- Comparative Analytics Dashboard on Port 8009

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves Reasoning Benchmark web UI
GET	/api/benchmark	Runs benchmark suite across all strategies and returns aggregated metrics

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

```
cd 09_reasoning_benchmark/src && python -m uvicorn server:app --host 127.0.0.1 --port 8009
```

Automated Test Suite Execution:

```
python -m pytest 09_reasoning_benchmark/tests/
```